

Referee package (DRAFT) – DR-2 for the lattice class: an unconditional, decoupling-free additive-energy bound on the sphere

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-10 · this version issued 2026-06-10 · v1.0

1. Purpose and scope

This referee document concerns DR-2 (the additive-energy / discrete-restriction bound) for the lattice competitor class. It is a harmonic-analysis / additive-combinatorics result, of interest independently of TECT. **Scope:** $Q = \Lambda \cap \{|x|^2 = R\}$ a fixed crystallographic lattice shell. **Not claimed:** the arbitrary (non-lattice) Q case remains OPEN (the decoupling-based route is T4 conditional).

2. The result

For $Q = \Lambda \cap \{|x|^2 = R\}$ (Λ a fixed crystallographic lattice, rational after clearing denominators), the additive energy obeys

$$E_+(Q) \leq (1 + 6 \max_{m \neq 0} d(4R - |m|^2)) N^2 \leq (1 + C_\varepsilon R^\varepsilon) N^2,$$

unconditionally and decoupling-free, where $d(\cdot)$ is the divisor function. For Gauss-typical shells $R \sim N^2$ this gives $E_+ \leq_\varepsilon N^{2+\varepsilon}$.

3. Structure of the argument

By Lemma A (sum-circle richness), $E_+(Q) \leq (1 + T'(Q)) N^2$ with T' the maximal proper sum-circle occupancy. The substitution $y = 2x - m$ sends $\mathbb{Z}^3 \cap C_m$ injectively into a homogeneous rank-2 representation count $\leq 6 d(4R - |m|^2)$ (Dirichlet class-number formula, uniform in m), $\ll_\varepsilon R^\varepsilon$ by the divisor bound. Hence $T' \ll_\varepsilon R^\varepsilon$ and the stated bound follows – no ℓ^2 -decoupling is used.

4. Numerical reproduction

Seven scripts reproduce the circle-richness lemma, the divisor-circle proof, the lattice-divisor scaling, the decoupling-iteration cross-checks and the cross-energy lemma; each carries asserts. Run them from the bundle.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.

(β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.

(γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / DR-2 referee package (DRAFT)
Precise statement:	For $Q=\text{Lambda cap } \{ x ^2=R\}$ (fixed crystallographic lattice), $E_+(Q) \leq (1+6 \max_{\{m \neq 0\}} d(4R- m ^2))N^2 \leq (1+C_{\text{eps}} R^{\text{eps}})N^2$, unconditional and decoupling-free, via Lemma A + the homogeneous representation count (Dirichlet class-number formula) + divisor bound. Gauss-typical $R \sim N^2$ gives $E_+ \leq_{\text{eps}} N^{2+\text{eps}}$.
Scope:	lattice shells $Q=\text{Lambda cap } \{ x ^2=R\}$. T7 modulo textbook number theory; arbitrary- Q OPEN.
Evidence grade:	unconditional lattice DR-2 (T7 mod NT); EXECUTED (7 scripts PASS).
Reproduction command:	<code>python codes/vacuum/dr2_circle_richness.py ; dr2_divcirc_proof.py ; dr2_lattice_divisor.py ; ...</code>
Falsifier:	a lattice shell with E_+ exceeding the divisor bound; or a counterexample to the rank-2 representation count.
Tier before / after:	DRAFT referee package; lattice DR-2 T7 (mod NT). NOT operator-confirmed.
No-overclaim statement:	unconditional for the LATTICE class (modulo textbook number theory); arbitrary non-lattice Q is OPEN.
Next required action:	operator review + confirm -> PUBLISHED bundle.